AgentPulse

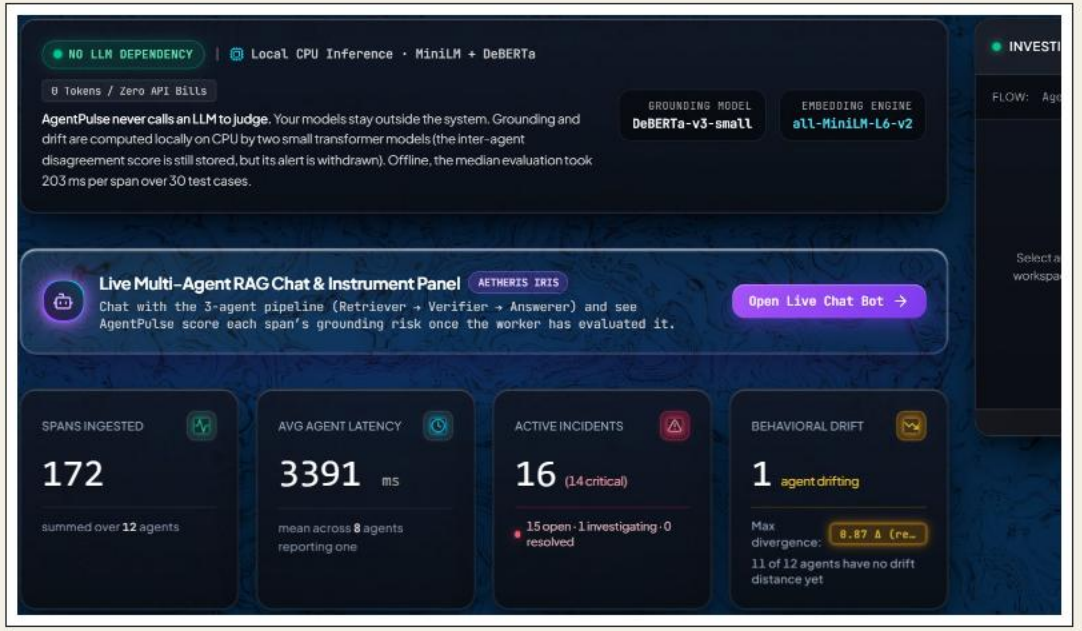
Observability for multi-agent LLM pipelines

LIVE EDITION: AGENTPULSE.CENTRALINDIA.CLOUDAPP.AZURE.COM CODE: GITHUB.COM/SOUM-CODE/AGENTPULSE

Every agent step, checked for grounding and drift, without an LLM judge

Two small local models score each span. Two signals held up on real data, two did not, and this paper says which.

BY SOMNATH REDDY



The deployed console on 1 November 2026. PHOTO: SCREENSHOT OF THE LIVE SITE

THE PROBLEM

In a multi-agent pipeline, a wrong answer looks exactly like a right one



Every call returns 200, nothing throws, latency is normal. Schematic, not data.

Grounding failure: an agent says what its evidence does not support.

Drift: an agent slowly answers differently after a change.

Most tools trace cost and latency; quality checks are optional and sampled.

SOURCE *README.md, introduction (tools treat quality evaluation as an optional, sampled add-on).*

THE SYSTEM

AgentPulse scores every span on a CPU, off the agent's critical path



No LLM call anywhere in the evaluator. The SDK adds 0.005 ms per call and never blocks the agent.

SOURCE *README.md, Architecture in short; HOW_IT_WORKS.md (bounded buffer, cooldown 900 s); benchmarks/benchmark_results.json, 2026-08-18 (0.005 ms, P50 over 2,000 calls).*

METHOD · GROUNDING

Grounding risk is an NLI probability, and its 0.3 weight is a design choice

```
risk = P(contradiction) + 0.3 × P(neutral)
```

LOW · 0 TO 0.4 | MEDIUM · ABOVE 0.4 | HIGH · ABOVE 0.7

CONTRADICTED | **UNVERIFIABLE** | **SUPPORTED**

Verdict = whichever of the three probabilities is highest.

DeBERTa compares each output with the evidence the agent was given.

Unverifiable: neither supported nor contradicted. A status, not a pass.

Weights 0 to 0.3 give identical results on our data.

SOURCE *GROUNDING_RECALIBRATION_2026-10.md, section 3.6 (analysis 2026-10-22); HOW_IT_WORKS.md, the verdict rule; backend/app/services/grounding.py (NEUTRAL_RISK_WEIGHT = 0.3).*

RESULT · GROUNDING

On 68 hand-labelled cases, grounding catches contradictions with few false alarms

EVALUATION SET	CASES	F1	FALSE ALARMS
Test split	30	0.963	0.059
Held-out pool, cut-off 0.7	47	0.927	0.074
Held-out pool, cut-off 0.4	47	0.952	0.074

False alarms = share of supported cases flagged (1 of 17 on the test split; 2 of 27 held out).

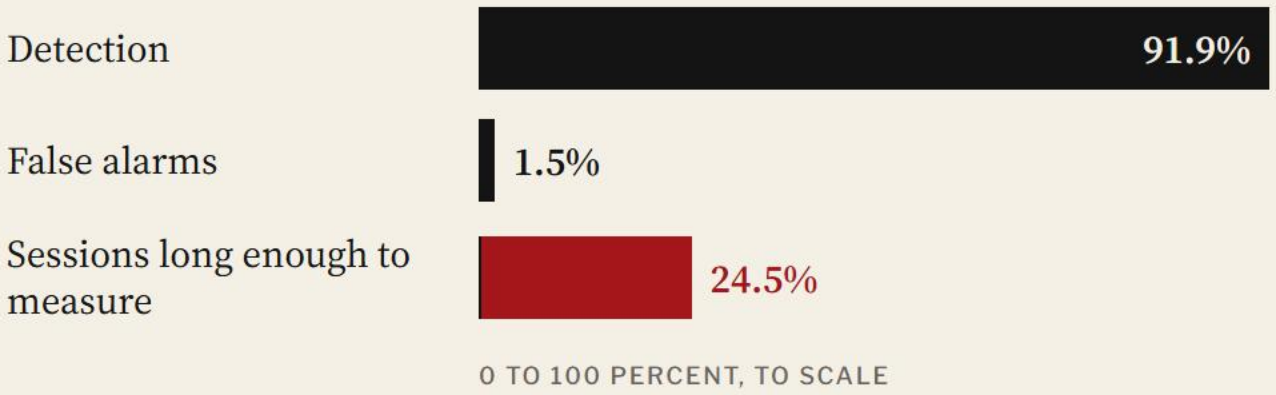
A small sample, not production traffic.

68 cases: 39 supported, 29 failures. 5 of the 30 test cases copy development cases.

SOURCE *GROUNDING_RECALIBRATION_2026-10.md, tables T3c and T3d (analysis 2026-10-22); README.md, Numbers and where they come from.*

RESULT · DRIFT

Drift detection is accurate, but only a quarter of real sessions are long enough to measure



Baseline: an agent's first 20 outputs.
Window: its latest 12.

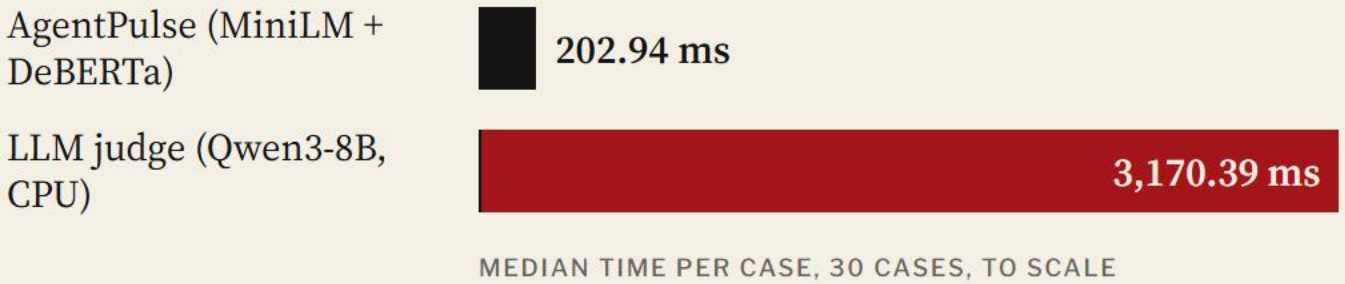
Alert when the two drift apart by a cosine distance above **0.30**.

Offline, real-trace corpus, held-out split, measured once.

SOURCE *DRIFT_REAL_TEXT_DIAGNOSIS_REPORT.md, sections 11.3 and 11.5 (2026-08-27); HOW_IT_WORKS.md (baseline 20, window 12, threshold 0.30).*

RESULT · COST

The evaluator is 15.6 times faster than an LLM judge, at a small accuracy cost



Same 30 cases: judge F1 **1.000**, AgentPulse **0.963**.

One judge model on one machine; not a claim about every judge.

SOURCE *LLM_JUDGE_COMPARISON_REPORT.md, median latency table and correction banner (2026-08-26; F1 at the shipped weight derived from stored probabilities).*

RESULT · VERIFICATION

The live site passed an end-to-end test from outside on 27 October

CHECK ON THE LIVE SITE	RESULT
Health endpoints	200
Key routes, writes without a key	401
Chat turn, three agents on Groq	6.35 s, all scored
Lab hallucination run	HIGH alerts
Every console view, two palettes	0 NaN

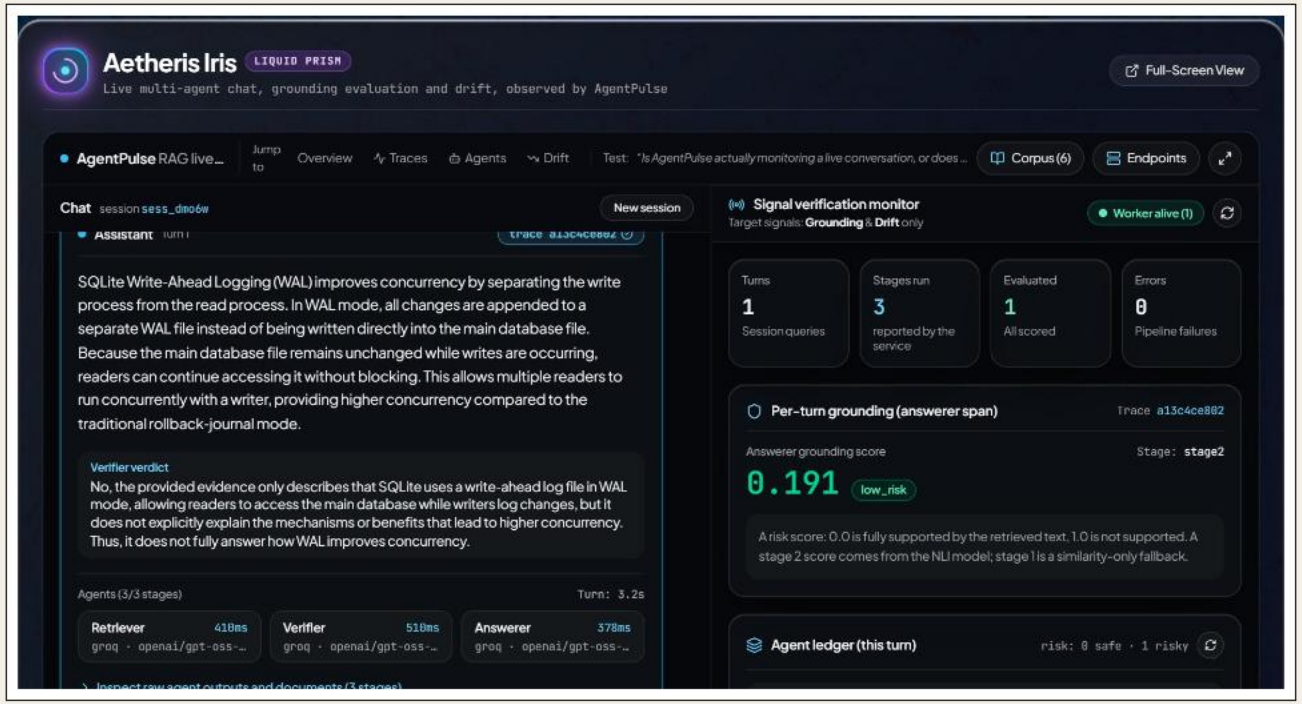
628 backend and **404** dashboard tests pass.

The test also found two real bugs; both were fixed.

SOURCE *docs/E2E_TEST_2026-10-27.md, items 1-25 and findings F1-F3; README.md, Testing and benchmarks (re-run 2026-10-27).*

LIVE DEMO · AGENTPULSE.CENTRALINDIA.CLOUDAPP.AZURE.COM

One question, three agents, and every step scored as it happens



A real turn on 1 November: 3.2 s, all agents on Groq; answerer risk 0.191. One turn, not a rate. PHOTO: LIVE SITE

1. Ask how SQLite WAL improves concurrency
2. Read the three stages
3. Open the agent ledger
4. Ask an out-of-corpus question
5. Export the trace (key X)

SOURCE *DEMO_GUIDE.md, steps 1-5; screenshot of the live RAG Live view, 1 November 2026.*

ENGINEERING

One small Azure VM, deployed automatically and closed to writes

DEPLOYMENT

One VM, five containers

Azure B2als_v2: 2 vCPU, 4 GB. Docker Compose with TLS.

CI / CD

Tested, then shipped

Every pull request runs the tests. A push to main deploys and rolls back if unhealthy.

SECURITY

Read-open, write-closed

Writes need a password. Key routes return 401. Every public route is rate-limited.

SOURCE *deploy/azure/README.md (VM size); .github/workflows/tests.yml, dashboard.yml, deploy-azure.yml; deploy/azure/Caddyfile; README.md, What works live; SESSION_HANDOFF.md 29.17.2.*

HONEST LIMITS

Two of the four signals failed on real traces, and the evidence base is small

LIMIT	EVIDENCE
Inter-agent disagreement	0 of 10 found
Tool-claim check	F1 0.000
Labelled grounding cases	68
Sessions long enough for drift	24.5%
Deployment	1 VM, 1 SQLite file

Grounding and drift are the two signals that held up.

The two failed alerts are withdrawn, not hidden.

SOURCE *README.md, Known limits; COMPETITIVE_POSITIONING.md 5.2; TOOL_CLAIM_EXTERNAL_TEST_REPORT.md section 10; GROUNDING_RECALIBRATION_2026-10.md; DRIFT_REAL_TEXT_DIAGNOSIS_REPORT.md 11.5.*

CONCLUSIONS

Conclusions

1. Every span can be scored on a CPU without an LLM judge: **202.94 ms** per case.
2. Grounding (F1 **0.963**) and drift held up; two other signals failed and are withdrawn.
3. It is live, tested (**628 + 404**) and open to examiners.

Next: more labelled data, a server database, test-gated deploys.

SOMNATH REDDY

`agentpulse.centralindia.cloudapp.azure.com`
`github.com/Soum-Code/agentpulse`

SOURCE *LLM_JUDGE_COMPARISON_REPORT.md; GROUNDING_RECALIBRATION_2026-10.md T3c; README.md, Testing and benchmarks (2026-10-27).*

APPENDIX · THE 0.3 WEIGHT

After a label-order bug was fixed, 0.5 flagged 7 of 17 correct answers; 0.3 flags 1

NEUTRAL WEIGHT	F1	FALSE ALARMS
0.0 to 0.3	0.963	0.059
0.4	0.897	0.176
0.5	0.788	0.412

Test split, cut-off 0.4, corrected labels. At the 0.7 cut-off every weight up to 0.6 gives the same result.

Before 21 October the code read two DeBERTa labels in swapped order. PR #69 fixed it; every figure was re-measured.

SOURCE *GROUNDING_RECALIBRATION_2026-10.md, banner and findings 4-7 (T3c, T5b, T7); PR #69; PR #89.*

APPENDIX · NEGATIVE RESULTS

Both withdrawn signals looked good on hand-written cases and failed on real traces

SIGNAL	HAND-WRITTEN	REAL TRACES
Tool-claim check, F1	0.842	0.000
Inter-agent disagreement	—	0 of 10 found

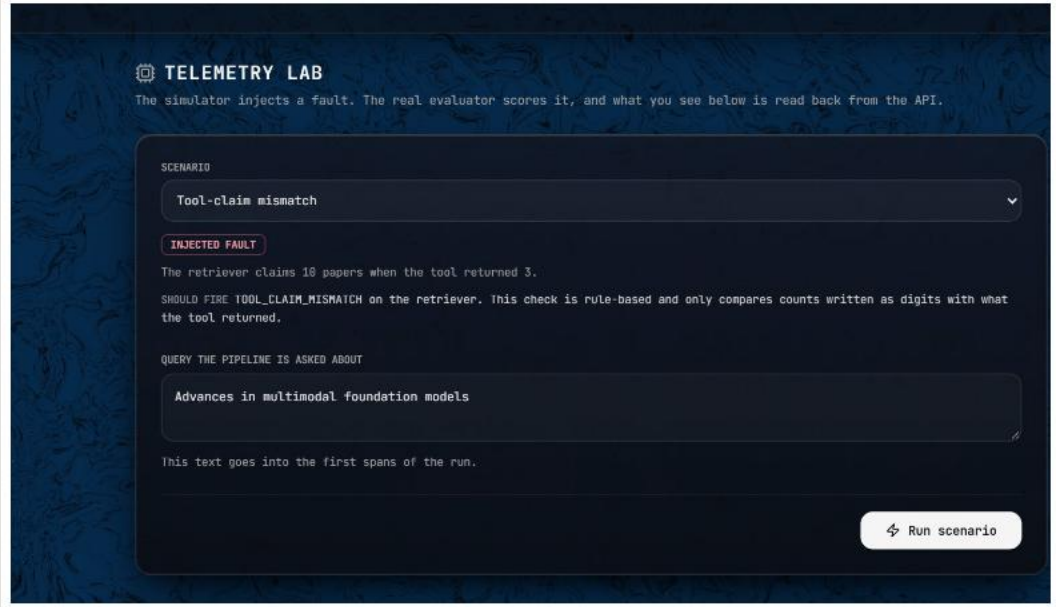
Tool-claim: 19 hand-written cases; no claim found in 8,353 real prose spans. Disagreement: AUC 0.457 without the planner, which is chance.

Real agents write counts in words, not digits; the tool-claim check only reads digits.

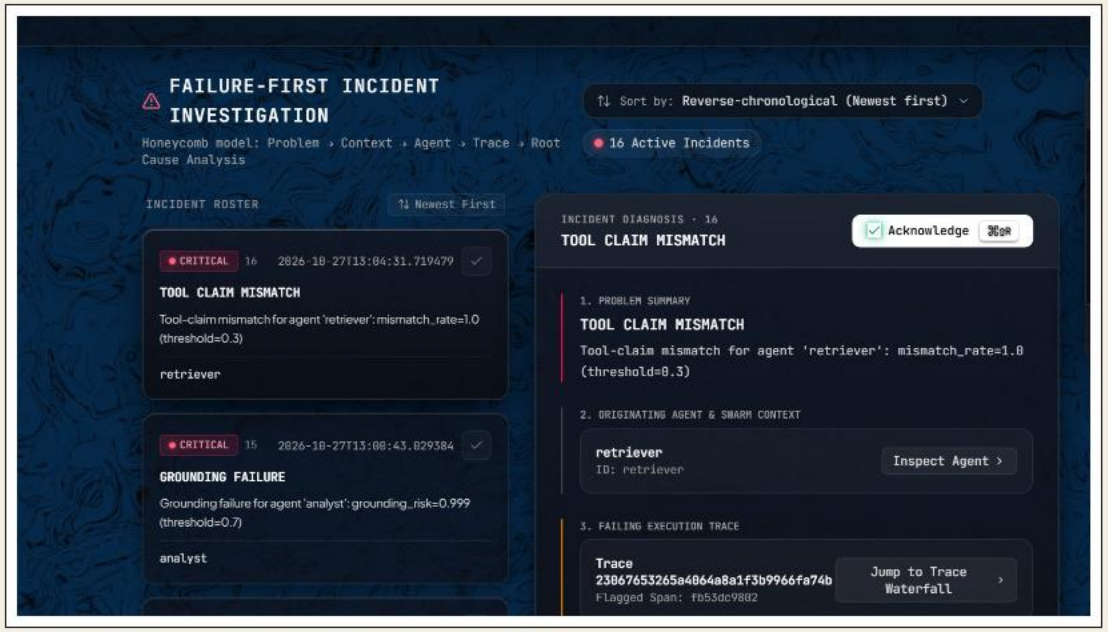
SOURCE *TOOL_CLAIM_EXTERNAL_TEST_REPORT.md section 10; COMPETITIVE_POSITIONING.md 5.1, 5.2; SESSION_HANDOFF.md 14.1, 24.9.5.*

APPENDIX · TELEMETRY LAB

A fault injected in the Lab becomes a scored trace and a HIGH incident



Five scenarios; the agent text is fixed, the evaluator is real. PHOTO: LIVE SITE



Lab runs of 27 October: GROUNDING_FAILURE for analyst, risk 0.999. PHOTO: LIVE SITE

SOURCE *HOW_IT_WORKS.md, Telemetry Lab; docs/E2E_TEST_2026-10-27.md, items 11, 12 and 22.*

REFERENCES

Every figure in this deck comes from these documents in the repository

README.md (state of 2026-10-27)
docs/E2E_TEST_2026-10-27.md
GROUNDING_RECALIBRATION_2026-10.md
DRIFT_REAL_TEXT_DIAGNOSIS_REPORT.md
LLM_JUDGE_COMPARISON_REPORT.md
TOOL_CLAIM_EXTERNAL_TEST_REPORT.md
COMPETITIVE_POSITIONING.md

HOW_IT_WORKS.md; DEMO_GUIDE.md
benchmarks/benchmark_results.json
experiments/results/ablation_results.json
deploy/azure/README.md; .github/workflows/
SESSION_HANDOFF.md, sections 14, 24, 29
Models: sentence-transformers/all-MiniLM-L6-v2; cross-encoder/nli-deberta-v3-small

SOURCE *Repository github.com/Soum-Code/agentpulse; model names from HOW_IT_WORKS.md.*